

Abu Jihad Arafat

PhD Applicant · Autonomous Systems · Robust Perception · Human-in-the-Loop and Simulation-Based Systems

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RESEARCH INTERESTS

Seventeen years of flying and instructing in safety-critical aviation, including flying in poor visibility and teaching others to do it. That work is why I study systems that have to keep working when the input degrades and when a human is inside the loop.

- **Robust machine perception in degraded visual environments.** Object detection and perception-difficulty estimation under fog, rain and low light, for autonomous driving and aerial robotics. This is the subject of my MSc thesis.
- **Safe decision-making for autonomy.** Extending perception towards state estimation, planning and control for unmanned aerial systems.
- **Human behaviour in simulated and safety-critical environments.** How people learn, adapt and fail at spatial and time-pressured tasks, studied through simulation, virtual reality and structured assessment.

EDUCATION

M.Sc. in Data Science & Machine Learning

2025 – 2026 (expected)

State University of Bangladesh · 40 credits · **CGPA 4.00 / 4.00 (expected)**

Thesis defence expected September 2026.

Postgraduate Diploma, Aviation Instruction

2021 – 2023

Bangladesh University of Professionals · 60 credit hours · CGPA 3.53 / 4.00

B.Sc. in Aeronautics

2008 – 2010

Bangladesh University of Professionals · CGPA 3.47 / 4.00, First Division · distinction in Aero Engine and Airmanship

MSc coursework: Machine Learning for Data Science; Artificial Intelligence for Data Science; Big Data Analysis; Programming for Data Science; Linear Algebra and Statistics; Applied Regression Analysis and Time-Series Forecasting; Advanced Sampling Methods and Statistical Inference; Multivariate Analysis; Data Wrangling and Visualization; Database Management Systems; Bioinformatics.

MSC THESIS RESEARCH

Perception-Difficulty Estimation in Degraded Visual Environments

in progress

Supervisor: Prof. Dr. Pintu Chandra Shill, Head, Dept. of CSE, State University of Bangladesh. Code and figures: github.com/aaaraafaat

- Testing whether **training-free, physics-based image cues** (dark-channel prior, saturation, contrast, entropy and others) can predict how hard a given image will be for an object detector, without any enhancement or restoration step first.
- Built a reproducible feature-extraction and data-cleaning pipeline on the RTTS real-fog dataset (4,322 images). The emphasis is on where each cue fails, including condition-dependent failure modes in which night scenes and coloured light mislead the cue in a systematic way.
- The framework is designed to extend to other degradations such as rain and to other sensing modalities. Stack: Python, PyTorch, Ultralytics/YOLO, OpenCV.

RESEARCH & DEVELOPMENT PROJECTS

Applied R&D in training technology, simulation and systems integration, Bangladesh Air Force. Portfolio: github.com/aaaraafaat

- **Flight-training simulator (PT-6 / PC-based AVTD), 2024–2025 — project lead.** Integrated off-the-shelf hardware with flight-simulation software (Prepar3D, MSFS) and a custom fixed-wing model, to give ab-initio and part-task pilot training at low cost. Trainee performance in the simulator was measured and used to adjust the syllabus.
- **VR flight simulator, from 2018.** Delivered the concept for a virtual-reality part-task trainer and presented it to the Chief of Air Staff. Later consulted on the build of a VR-based PCATD for combat simulation, which was completed and put into use.
- **Automated flight-mission scheduling and progress-monitoring tool, 2025–2026.** Web application that builds daily flying schedules against aircraft, instructor and trainee constraints, and tracks each trainee's progress through the syllabus.
- **Automated map-preparation software, 2019 — functional manager.** Google Maps API with layered custom charts, to automate the preparation of mission route and navigation charts.

PROFESSIONAL EXPERIENCE

Project Manager — Air Warfare Centre, Cyberwarfare & IT Directorate

Dec 2025 – present

Bangladesh Air Force

- Project manager for a **USD 5.2M training-integration programme:** contract terms, contractor and technology integration, security and delivery readiness.

Flying Instructor, Flight Commander and Training Operations Manager

Mar 2022 – Dec 2025

Bangladesh Air Force Academy, then Flying Instructors' School, Bogura

- Taught **65 cadet pilots and 26 instructor pilots** in the air and on the ground. Designed and ran the assessments, check-rides and technical briefings used to grade them.
- Tracked how individual trainees learned and where they plateaued, and used that to change how a sortie was taught.
- Planned and ran daily flight-training operations. At the Academy the trainees were cadets. At the Flying Instructors' School they were instructor pilots, and the work included supervising and planning the instructor training programme.
- **Operational Manager for the G120TP training systems**, authorised by Grob Aircraft. Ran the mission planning and debriefing system, the ground-based training system, the cockpit procedures trainer and the flight simulation training device.

- Built a Google Sheets progress tracker with automated cross-sheet calculations, and a standalone web app for visualising training progress and building schedules.

Staff Officer, Aviation Safety — UN Hybrid Operation in Darfur (UNAMID)

2015 – 2016

United Nations Peacekeeping Mission, Sudan

- Ran the mission aviation-safety programme in an active conflict zone. Prepared the monthly, quarterly and annual analysis reports and the executive dashboards used by mission leadership.

Earlier (2009–2015): operational fighter pilot, F-7BG, 1,278 total flight hours. Staff appointments including Base Adjutant, coordinating administration for more than 400 personnel.

TEACHING & ASSESSMENT

Adjunct Faculty

2023 – 2025

Air Law (PG Diploma, 3 semesters) · Airmanship (B.Sc. Aeronautics, 2 semesters). Also lectured Air Law and Instructional Technique on the Flying Instructors' Course.

Assessor, Inter Services Selection Board

2019

Group Testing Officer, ISSB Bangladesh

- Assessed candidates on 10 live officer-selection boards. My ratings fed the selection or rejection decision. The method is structured behavioural observation, not a written test: candidates are put through standardised group and individual tasks such as planning exercises, command tasks and obstacle problems, and the assessor records how they decide, adapt and influence others under time pressure. Ratings are cross-checked against a psychologist's instruments and an interview board.

Aviation Safety Awareness Trainer — United Nations (UNAMID)

2016

Designed and delivered induction training for more than 500 mission personnel. Designed the e-learning content (Articulate LMS).

TECHNICAL SKILLS

- **Programming and machine learning** (applied in MSc research): Python, NumPy, pandas, OpenCV, PyTorch and the Ultralytics/YOLO ecosystem at an applied level. SciPy and scikit-learn for statistical work.
- **Computer vision and research methods:** object detection, image-degradation modelling (dark-channel prior), experiment design, statistical validation.
- **Simulation and human-in-the-loop:** Prepar3D and MSFS simulator integration, VR part-task trainer design, mission planning and debriefing systems, syllabus design and performance assessment.
- **Tools:** Git/GitHub, Google Colab, LaTeX/Overleaf, Zotero, Matplotlib, standalone HTML/JS dashboards.

CERTIFICATIONS & TRAINING

- **Qualified Flying Instructor** — Flying Instructors' School, Bangladesh Air Force. 24-week course, 2021–2022.
- **Flight Safety Officer** — Flight Safety Institute, Bangladesh Air Force, 2017. Grade A. Examined subjects included aviation psychology, aviation physiology and human factors in accident investigation.
- **Group Testing Officer (assessor)** — Inter Services Selection Board, Bangladesh. 11-week course, 2019.
- **Instructor type rating, Grob G120TP** (Grob Aircraft, Germany, 2022), and **operational-manager authorisation** for the G120TP mission planning, ground-based training and flight simulation systems (Grob Aircraft, 2022).
- **Operational Training in Aerospace Medicine** — Institute of Aerospace Medicine, Indian Air Force, Bangalore, 2015.
- **Aviation Risk Management** — United Nations, UNAMID Integrated Mission Training Centre and Department of Field Support, 2016.
- **Online certificate programmes:** IBM Data Science Professional; Google Cybersecurity Professional; IBM Project Manager.

HONOURS, LANGUAGES & TESTS

- **Birshrestha Matiur Rahman Trophy** — best performance in flying training, Bangladesh Air Force Academy. **United Nations Medal (UNAMID).** **Four Certificates of Appreciation** — base development, national earthquake-preparedness exercise, and contribution beyond peers.
- **TOEFL iBT 103 / 120**, December 2024 (Reading 24, Listening 26, Speaking 25, Writing 28). **Languages:** English (C1); Bangla (native).

REFERENCES

- **Prof. Dr. Pintu Chandra Shill** — Head, Department of CSE, State University of Bangladesh. MSc thesis supervisor.
- **Air Cdre Dr. Kamrul Huda, PhD** — Director, Cyberwarfare & IT Directorate, Bangladesh Air Force. Current reporting officer.

Full contact details available on request.